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Indian Standard

IS 4985 : 2021

जल की पूर्ति के लिए
असुघट्टित पी. वी. सी. पाईप — विशिष्टि
(चौथा पुनरीक्षण)

Unplasticized PVC Pipes for
Water Supplies — Specification
(*Fourth Revision*)

ICS 23.040.20; 91.140.60

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भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS
मानक भवन, 9 बहादुरशाह ज़फर मार्ग, नई दिल्ली – 110002
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI-110002
www.bis.gov.in www.standardsbis.in

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FOREWORD

This Indian Standard (Fourth Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Plastic Piping System Sectional Committee had been approved by the Civil Engineering Division Council.

This standard was first published in 1968 covering unplasticized polyvinyl chloride pipes (UPVC) pipes of sizes 16 to 315 mm. The standard was subsequently revised in 1981 for incorporating dimensions of bell or socket end pipes and pipes suitable for plumbing work in buildings. Provision of reversion test by the oven method as an alternative to the immersion method and additional test on resistance to sulphuric acid were also incorporated. Long-term and short-term hydraulic tests were replaced by internal hydrostatic pressure tests in line with ISO/DIS 4422 'Unplasticized polyvinyl chloride (PVC) pipes and fittings for water supply — Specification'. Sampling clauses were reviewed after carrying out a detailed study of the process of manufacture and statistically analyzing the data collected from the manufacturers of unplasticized PVC pipes. Later, through an amendment, the range of pipe sizes covered was increased up to 630 mm.

The second revision of this standard was brought out in 1988 for incorporating further changes made necessary in the light of the experience gained in the use of UPVC pipes in India and technological advancements in the manufacture of these pipes in India and abroad. The changes included additional test methods on determination of cadmium and mercury contents. The opacity test was modified in line with the ISO standard and an alternate test method for determination of opacity was included. Further, the Committee responsible for the formulation of that standard felt that the test methods should be published separately in parts to facilitate further review of each part as this would also be in line with ISO procedure. Accordingly, the various test methods were covered in a separate standard, namely IS 12235 (Parts 1 to 11) : 1986 'Methods of test for unplasticized PVC pipes for potable water supplies'.

Further technological advancements in this field advocated the inclusion of UPVC pipes with sockets for use with elastomeric sealing rings in its third revision which was brought out in 2000. Keeping in view the advent of globalization and the likelihood of exports, the standard was brought more in line with ISO 4422 'Pipes and fittings made of unplasticized poly (vinyl chloride) (PVC-U) for water supply — Specifications'. Test for determination of reversion, resistance to sulphuric acid and stress-relief test were deleted. Additional tests for specific gravity, vicat softening temperature and sulphated ash content test, were included. The impact resistance test was modified to be in line with ISO 3127 : 1994 'Thermoplastics pipes — Determination of resistance to external blows — Round-the-clock method' to the extent possible.

In the formulation of the third revision of the standard, considerable assistance was derived from the following International Standards:

ISO 161-1-1978	Thermoplastic pipes for the transport of fluids — Nominal outside diameters and nominal pressures — Part 1: Metric series
ISO/DIS 727-1985(E)	Fittings of unplasticized PVC, chlorinated PVC or ABS with plain sockets for pipes under pressure — Dimensions of sockets — Metric series
ISO 2045 : 1988	Single sockets for UPVC and CPVC pressure pipes with elastic sealing ring type joints — Minimum depths of engagement
ISO 3127 : 1994(E)	Thermoplastics pipes — Determination of resistance to external blows — Round-the-clock method
ISO 3603 : 1977	Fittings for unplasticized polyvinyl chloride pressure pipes with elastomeric sealing ring type joints — Pressure test for leak proofness
ISO 4422 : 1992	Unplasticized polyvinyl chloride pipes and fittings for water supply — Specifications

(Continued to third cover)

Indian Standard

**UNPLASTICIZED PVC PIPES FOR
WATER SUPPLIES — SPECIFICATION**

(Fourth Revision)

1 SCOPE

1.1 This standard covers requirements for plain as well as socket-ended pipes, including those for use with elastomeric sealing rings, for potable water supplies.

1.1.1 The standard also cover pipes for agricultural use.

1.2 This standard does not cover unplasticized PVC pipes for use in suction and delivery lines of agricultural pumps, which have been covered in IS 12231.

1.3 The pipes covered in this standard are not suitable for use as casing pipes in tube wells. Such pipes are covered in IS 12818.

2 REFERENCES

The standards listed in Annex A contain provisions which, through reference in this text, constitute provision of this standard. At the time of publication the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate, the possibility of applying the most recent editions of the standards.

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Nominal Size (DN) — The numerical designation for the size of a pipe, other than a pipe designated by thread size, which is a convenient round number approximately equal to the manufacturing dimension in millimetres (mm).

3.2 Nominal Outside Diameter (d_n) — The specified outside diameter, in millimetres assigned to a nominal size.

3.3 Outside Diameter at any Point (d_o) — The value of the measurement of the outside diameter of a pipe through its cross-section at any point of the pipe, rounded off to the next higher 0.1 mm.

3.4 Mean Outside Diameter (d_{em}) — The quotient of the outer circumference of a pipe and 3.142 (π) in any cross-section, rounded off to the next higher 0.1 mm.

3.5 Minimum Mean Outside Diameter (d_{em}, Min) — The minimum value for the mean outside diameter as specified for a given nominal size.

3.6 Maximum Mean Outside Diameter (d_{em}, Max) — The maximum value for the mean outside diameter as specified for a given nominal size.

3.7 Mean Inside Diameter at Mid Point of Socket Length (d_{im}) — The arithmetical mean of two measured inside diameters perpendicular to each other at the mid-point of the socket length.

3.8 Out-of-Roundness (Ovality) — The difference between the measured maximum and the measured minimum outside diameter in the same cross-section of the pipe.

3.9 Nominal Wall Thickness (e_n) — A numerical designation of the wall thickness of a component which is a convenient round number, approximately equal to the manufacturing dimension in millimetres (mm).

3.10 Wall Thickness at any Point (e) — The value of the measurement of the wall thickness at any point around the circumference of a pipe, rounded off to the next higher 0.1 mm.

3.11 Minimum Wall Thickness at any Point (e_{min}) — The minimum value for the wall thickness at any point around the circumference of a pipe, rounded off to the next higher 0.1 mm.

3.12 Maximum Wall Thickness at any Point (e_{max}) — The maximum value for the wall thickness at any point around the circumference of a pipe, rounded off to the next higher 0.1 mm.

3.13 Mean Wall Thickness (e_m) — The arithmetical mean of at least four measurements regularly spaced around the circumference and in the same cross-section of a pipe, including the measured minimum and the measured maximum values of the wall thickness in that cross-section and rounded off to the next higher 0.1 mm.

3.14 Tolerance — The permitted variation of the specified value of a quantity, expressed as the difference between the permitted maximum and the permitted minimum value.

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3.15 Working Pressure (PN) — The numerical designation of a pipe related to the mechanical characteristics of that pipe used for reference purposes. For plastic piping systems, it corresponds to the allowable operating pressure, in bar, conveying water at 27 °C.

3.16 Allowable Operating Pressure (PFA) — The maximum hydrostatic pressure, excluding surge, which is allowed in continuous use with water within the temperature range concerned. It is calculated using the following equation:

$$PFA = f_T \times PN$$

where

f_T = derating factor depending on water temperature; and

PN = working pressure.

NOTE — In cases where a further derating (or uprating) factor depending on the application is required,

$$PFA = f_A \times f_T \times PN$$

where

f_A = factor depending on the application.

3.17 Hydrostatic Stress (σ) — The stress induced in the wall of a pipe when a pressure is applied using water as a medium. The hydrostatic stress is related to the applied pressure, P , the wall thickness at any point, e , and the mean outside diameter, d_{em} , of a pipe and calculated using the following approximation equation:

$$\sigma = \frac{P(d_{em} - e)}{2e}$$

where σ and P are in same units.

3.18 Long-Term Hydrostatic Stress — The constant hydrostatic stress that is maintained during a sustained period of time.

3.19 Socket-Ended Pipe — Unplasticized PVC pipes whose one end is expanded after heating for the purpose of jointing by solvent cement or jointing using an elastomeric sealing ring, to the plain ends of unplasticized PVC pipes.

3.20 Tests

3.20.1 Type Tests — Tests carried out whenever a change is made in the composition or in the size/series in order to establish the suitability and the performance capability of the pipes.

3.20.2 Acceptance Tests — Tests carried out on samples taken from a lot for the purpose of acceptance of the lot.

3.21 Virgin Material — Material in such form as granules or powder that has not been subjected to use or processing other than that required for its manufacture and to which no reprocessable or recyclable material(s) have been added.

3.22 Own Rework Material — Material prepared from rejected unused pipes, including trimmings from the production of pipes, that will be reprocessed in a manufacturer's plant by a process such as extrusion and for which the complete formulation is known.

4 NOTATION

The following notations (symbols) shall apply in this standard:

d_n	=	Nominal outside diameter
d_e	=	Outside diameter at any point
d_{em}	=	Mean outside diameter
d_{em}, Max	=	Maximum mean outside diameter
d_{em}, Min	=	Minimum mean outside diameter
d_{im}	=	Mean inside socket diameter at midpoint of socket length
DN	=	Nominal size
e	=	Wall thickness at any point
e_m	=	Mean wall thickness
e_{max}	=	Maximum wall thickness at any point
e_{min}	=	Minimum wall thickness at any point
e_n	=	Nominal wall thickness
L_o	=	Overall length of pipe
L_e	=	Effective length of pipe
L_s	=	Minimum socket length
PN	=	Nominal pressure (working pressure)
f_A	=	Derating (or uprating) factor for application
f_T	=	Derating factor for water temperatures
ρ	=	Material density
G	=	Hydrostatic stress
σ_s	=	Design stress

5 CLASSIFICATION OF PIPES

5.1 The pipes shall be classified by pressure ratings (working pressures) at 27 °C as follows:

<i>Class of Pipe</i>	<i>Working Pressure (PN)</i>
Class 1	0.25 MPa (2.5 kg/cm ²)
Class 2	0.4 MPa (4.0 kg/cm ²)
Class 3	0.6 MPa (6.0 kg/cm ²)
Class 4	0.8 MPa (8.0 kg/cm ²)
Class 5	1.0 MPa (10.0 kg/cm ²)
Class 6	1.25 MPa (12.5 kg/cm ²)

NOTE — The above pipes are recommended for water temperatures ranging from +1 to +45 °C. The recommended maximum safe working stress for these pipes is 8.6 MPa at 27 °C. At higher temperature up to 45 °C, the strength of the pipe reduces and the working pressure shall be modified in accordance with Fig. 1. Occasional rise in temperature as in summer season with concurrent corresponding reduction in temperature during nights has no deleterious effect on the life/working pressure of the pipes considering the total life of pipes.

5.2 The pipes shall be classified based on their application as follows:

- Type A* — Pipes for water supply; and
- Type B* — Pipes for agricultural use.

NOTE — The pipes shall be marked for their identification for applicable use/application (see 13.1.2).

6 COMPOSITION

6.1 The material from which the pipe is produced shall consist substantially of unplasticized polyvinyl chloride to which may be added only those additives that are needed to facilitate the manufacture of the pipe and the production of sound and durable pipe of good surface finish, mechanical strength and opacity under conditions of use. None of these additives shall be used separately or together in quantities sufficient to constitute a toxic, organoleptic or microbial growth hazard, or materially to impair the fabrication or welding properties of the pipe, or to impair its chemical and physical or mechanical properties (in particular long-term mechanical strength and impact strength) as defined in the standard. The additives to be used shall be selected from IS 10148 and shall be uniformly dispersed.

6.1.1 The monomer content (VCM content) in the resin shall be within the limits specified in 4.4.1 of IS 10151, when tested as per Annex A of IS 10151.

6.1.2 The composition shall be based on PVC resin having a K value of 64 or greater when tested in accordance with IS 4669.

NOTE — A test report or conformity certificate may be obtained from the resin manufacturer for the VCM content (see 6.1.1) and K value (see 6.1.2) of the resin being used, unless the same is tested in an independent laboratory. The frequency of this test report or conformity certificate shall be once in every three months.

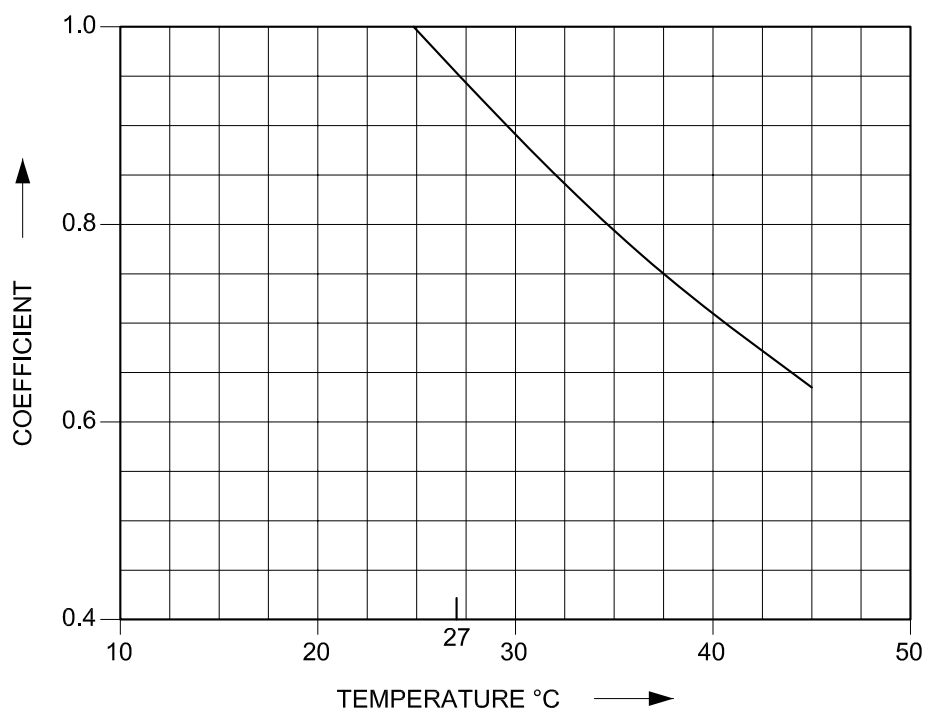


FIG. 1 GRAPH FOR MORE PRECISE CALCULATIONS GIVING THE MAXIMUM CONTINUOUS WORKING PRESSURE (COEFFICIENT) FOR TEMPERATURE UPTO 45 °C

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6.2 The addition of the manufacturer's own rework material is permissible. The quantity of the rework material used is to be declared by the manufacturer. No other rework material shall be used.

6.3 Conformity of pipes to this standard is tested by carrying out the tests specified in this standard. Pipes meeting the requirements of these tests are deemed to meet the requirements of the standard.

7 DIMENSIONS

7.1 Dimensions of Pipes

7.1.1 Diameters

The mean outside diameter, outside diameter at any point and their tolerances shall be as given in Table 1. This shall be measured according to the method given in IS 12235 (Part 1).

7.1.1.1 Mean outside diameters

The permissible variation ($d_{em} - d_n$) between the mean outside diameter (d_{em}) and the nominal outside diameter (d_n) of a pipe shall be positive in the form $+x$, where x is less than or equal to the greater of the following two values:

- 0.3 mm, and
- $0.003 d_n$ rounded off to the next higher 0.1 mm.

7.1.1.2 Diameter at any point

The permissible variation between the outside diameter at any point (d_e) and the nominal diameter (d_n) of a pipe (also called tolerance on ovality) shall not exceed the greater of the following two values:

- 0.5 mm, and
- $0.012 d_n$ rounded off to the next higher 0.1 mm.

7.1.2 Wall Thickness

The wall thickness of plain pipe (both for solvent cementing and elastomeric sealing ring joints) and the plain portion of socket ended pipe shall be as given in Table 1.

The above wall thickness shall be measured by any of the three methods given in 2.1.1 and 2.1.2 of IS 12235 (Part 1). To check the conformity of the wall thickness of the pipe throughout its entire length, it is necessary to measure the wall thickness of the pipe at any point along its length. This shall be done by cutting the pipe at any point along its length and measuring the wall thickness as above. Alternatively, to avoid destruction of the pipe, non-destructive testing methods such as the use of ultrasonic wall thickness measurement gauges shall be used at any four points along the length of the pipe.

7.1.2.1 Tolerance on wall thickness

- For pipes of minimum wall thickness 6 mm or less, the permissible variation between the minimum wall thickness (e_{min}) and the wall thickness at any point (e), ($e - e_{min}$) shall be positive in the form of $+y$, where $y = 0.1 e_{min} + 0.2$ mm.
- For pipes of minimum wall thickness greater than 6 mm, the permissible variation of wall thickness shall again be positive in the form of $+y$, where y would be applied in two parts as given at (c) and (d).
- The average wall thickness shall be determined by taking at least six measurements of wall thickness round the pipe and including both the absolute maximum and the absolute minimum values. The tolerance applied to this average wall thickness from these measurements shall be within the range $0.1 e_{min} + 0.2$ mm (see Table 1).
- The maximum wall thickness at any point shall be within the range $0.15 e$ (see Table 1).
- The results of these calculations for checking tolerance shall be rounded off to the next higher 0.1 mm.

7.1.3 The mean outside diameter, outside diameter at any point, and wall thickness of plumbing pipes when measured in accordance with IS 12235 (Part 1), shall be as given in Table 2.

Table 1 Dimensions of Unplasticized PVC Pipes

(Clauses 7.1.1 and 7.1.2)

All dimensions in millimetres.

Sl No.	Nominal Outside Diameter (Nominal Size)	Mean Outside Diameter	Outside Diameter at Any Point ^{b)}		Wall Thickness																		
					Class 1 0.25 MPa			Class 2 0.40 MPa			Class 3 0.60 MPa			Class 4 0.80 MPa			Class 5 1.00 MPa			Class 6 1.25 MPa			
			Min	Max	Min	Max	Avg	Min	Max	Avg	Min	Max	Avg	Min	Max	Avg	Min	Max	Avg	Min	Max		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)
i)	20	20.0	20.3	19.5	20.5													1.5	1.1	1.5	1.8	1.4	1.8
ii)	25	25.0	25.3	24.5	25.5										1.6	1.2	1.6	1.8	1.4	1.8	2.1	1.7	2.1
iii)	32	32.0	32.3	31.5	32.5										1.9	1.5	1.9	2.2	1.8	2.2	2.7	2.2	2.7
iv)	40	40.0	40.3	39.5	40.5							1.8	1.4	1.8	2.2	1.8	2.2	2.7	2.2	2.7	3.3	2.8	3.3
v)	50	50.0	50.3	49.4	50.6							2.1	1.7	2.1	2.8	2.3	2.8	3.3	2.8	3.3	4.0	3.4	4.0
vi)	63	63.0	63.3	62.2	63.8				1.9	1.5	1.9	2.7	2.2	2.7	3.3	2.8	3.3	4.1	3.5	4.1	5.0	4.3	5.0
vii)	75	75.0	75.3	74.1	75.9				2.2	1.8	2.2	3.1	2.6	3.1	4.0	3.4	4.0	4.9	4.2	4.9	5.9	5.1	5.9
viii)	90	90.0	90.3	88.9	91.1	1.7	1.3	1.7	2.6	2.1	2.6	3.7	3.1	3.7	4.6	4.0	4.6	5.7	5.0	5.7	7.0	6.1	7.1
ix)	110	110.0	110.4	108.6	111.4	2.0	1.6	2.0	3.0	2.5	3.0	4.3	3.7	4.3	5.6	4.9	5.6	7.0	6.1	7.1	8.5	7.5	8.7
x)	125	125.0	125.4	123.5	126.5	2.2	1.8	2.2	3.4	2.9	3.4	5.0	4.3	5.0	6.4	5.6	6.4	7.8	6.9	8.0	9.6	8.5	9.8
xi)	140	140.0	140.5	138.3	141.7	2.4	2.0	2.4	3.8	3.2	3.8	5.5	4.8	5.5	7.2	6.3	7.3	8.7	7.7	8.9	10.7	9.5	11.0
xii)	160	160.0	160.5	158.0	162.0	2.8	2.3	2.8	4.3	3.7	4.3	6.2	5.4	6.2	8.2	7.2	8.3	9.9	8.8	10.2	12.2	10.9	12.6
xiii)	180	180.0	180.6	177.8	182.2	3.1	2.6	3.1	4.9	4.2	4.9	7.0	6.1	7.1	9.0	8.0	9.2	11.1	9.9	11.4	13.7	12.2	14.1
xiv)	200	200.0	200.6	197.6	202.4	3.4	2.9	3.4	5.3	4.6	5.3	7.7	6.8	7.9	10.0	8.9	10.3	12.3	11.0	12.7	15.2	13.6	15.7
xv)	225	225.0	225.7	222.3	227.7	3.9	3.3	3.9	6.0	5.2	6.0	8.6	7.6	8.8	11.2	10.0	11.5	13.9	12.4	14.3	17.1	15.3	17.6
xvi)	250	250.0	250.8	247.0	253.0	4.2	3.6	4.2	6.5	5.7	6.5	9.6	8.5	9.8	12.6	11.2	12.9	15.4	13.8	15.9	18.9	17.0	19.6
xvii)	280	280.0	280.9	276.6	283.4	4.8	4.1	4.8	7.3	6.4	7.4	10.7	9.5	11.0	14.0	12.5	14.4	17.2	15.4	17.8	21.1	19.0	21.9
xviii)	315	315.0	316.0	311.2	318.8	5.3	4.6	5.3	8.2	7.2	8.3	12.0	10.7	12.4	15.6	14.0	16.1	19.3	17.3	19.9	23.8	21.4	24.7
xix)	355	355.0	356.1	350.7	359.3	5.9	5.1	5.9	9.2	8.1	9.4	13.4	12.0	13.8	17.6	15.8	18.2	21.8	19.6	22.6	26.8	24.1	27.8

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Table 1 (Concluded)

Sl No.	Nominal Outside Diameter (Nominal Size)	Mean Outside Diameter	Outside Diameter at Any Point ¹⁾	Wall Thickness																			
				Class 1 0.25 MPa			Class 2 0.40 MPa			Class 3 0.60 MPa			Class 4 0.80 MPa			Class 5 1.00 MPa			Class 6 1.25 MPa				
				Avg Max	Min	Max	Avg Max	Min	Max	Avg Max	Min	Max	Avg Max	Min	Max	Avg Max	Min	Max	Avg Max	Min	Max		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)
xx)	400	400.0	401.2	395.2	404.8	6.6	5.8	6.7	10.3	9.1	10.5	15.1	13.5	15.6	19.8	17.8	20.5	24.4	22.0	25.3	30.2	27.2	31.3
xxi)	450	450.0	451.4	444.6	455.4	7.4	6.5	7.5	11.6	10.3	11.9	17.0	15.2	17.5	22.2	20.0	23.0	27.5	24.8	28.6	33.8	30.5	35.1
xxii)	500	500.0	501.5	494.0	506.0	8.2	7.2	8.3	12.8	11.4	13.2	18.8	16.9	19.5	24.8	22.3	25.7	30.5	27.5	31.7	37.5	33.9	39.0
xxiii)	560	560.0	561.7	553.2	566.8	9.2	8.1	9.4	14.3	12.8	14.8	21.0	18.9	21.8	27.6	24.9	28.7	34.1	30.8	35.5	42.0	38.0	43.7
xxiv)	630	630.0	631.9	622.4	637.6	10.3	9.1	10.5	16.1	14.4	16.6	23.7	21.3	24.5	31.0	28.0	32.2	38.4	34.7	40.0	47.2	42.7	49.2

NOTES

1 The table is based on metric series of pipe dimensions given in ISO 161-1 in respect of pipe dimensions and ISO/DIS 4422.

2 The wall thickness of pipes is based on a safe working stress of 8.6 MPa at 27 °C and the working pressure gets reduced at sustained higher temperatures. Occasional rise in temperature as in summer season with concurrent corresponding reduction in temperature during nights has no deleterious effect on the life/working pressure of the pipes considering the total life of pipes.

¹⁾ For all sizes of Classes 1, 2 and 3, this requirement need not to be satisfied as the ratio of minimum wall thickness to nominal outside diameter does not exceed 0.035 in these cases.

NOTES

1 The table is based on metric series of pipe dimensions given in ISO 161-1 in respect of pipe dimensions and ISO/DIS 4422.

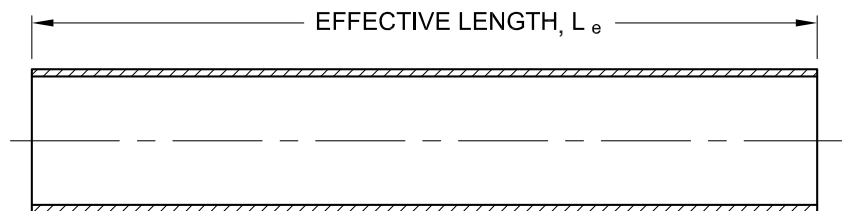
2 The wall thickness of pipes is based on a safe working stress of 8.6 MPa at 27 °C and the working pressure gets reduced at sustained higher temperatures. Occasional rise in temperature as in summer season with concurrent corresponding reduction in temperature during nights has no deleterious effect on the life/working pressure of the pipes considering the total life of pipes.

¹⁾ For all sizes of Classes 1, 2 and 3, this requirement need not to be satisfied as the ratio of minimum wall thickness to nominal outside diameter does not exceed 0.035 in these cases.

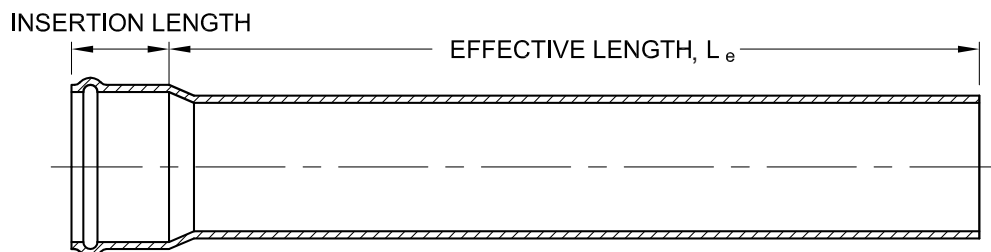
Table 2 Dimensions of UPVC Plain Ended Pipe for Plumbing in Buildings
(Clause 7.1.3)

All dimensions in millimetres.

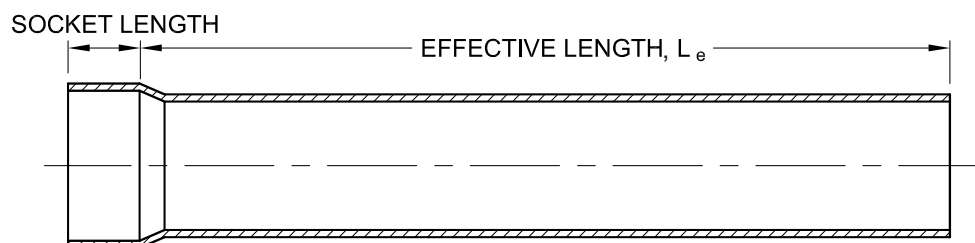
Sl No.	Nominal Outside Diameter d_n	Mean Outside Diameter d_{em}		Outside Diameter at Any Point d_e		Wall Thickness e	
		Min	Max	Min	Max	Min	Max
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	20	20.0	20.3	19.5	20.5	2.8	3.3
ii)	25	25.0	25.3	24.5	25.5	2.9	3.4
iii)	32	32.0	32.3	31.5	32.5	3.4	3.9
iv)	40	40.0	40.3	39.5	40.5	3.6	4.2
v)	50	50.0	50.3	49.4	50.6	3.7	4.3



2A PLAIN ENDED PIPE



2B SOCKETED PIPE FOR ELASTOMERIC SEALING RING JOINT



2C SOCKETED PIPE FOR SOLVENT CEMENT JOINTING

FIG. 2 EFFECTIVE LENGTHS OF PIPE

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7.1.4 Length

7.1.4.1 Effective length (L_e)

If the length of a pipe is specified, the effective length shall not be less than that specified. The preferred effective length of pipes shall be 4, 5, or 6 m. The pipes may be supplied in other lengths where so agreed upon between the manufacturer and the purchaser.

- Plain ended pipe* — For plain ended pipes, the overall length measured shall be the effective length as shown in Fig. 2A.
- Socketed pipe for elastomeric sealing ring jointing* — The effective length of such pipes shall be determined by subtracting from the overall length the insertion length as shown in Fig. 2B.
- Socketed pipe for solvent cement jointing* — The effective length of such pipes shall be determined by subtracting from the overall length the socket length as shown in Fig. 2C.

7.2 Dimensions of Sockets

7.2.1 Sockets formed on the ends of the pipes shall be reasonably parallel to the axis of the pipe.

7.2.1.1 Sockets for solvent cement jointing

When measured in accordance with the method given in IS 12235 (Part 1), the socket dimensions shall conform to dimensions given in Table 3 read along with Fig. 3.

The minimum length of any socket shall be given by the expression, $L_s = 0.5 d_n + 6$ mm,

where

L_s = minimum socket length, and

d_n = nominal outside diameter of the pipe.

NOTES

1 The mean inside diameter of the socket is defined as the arithmetical mean of two diameters measured at 90° to each other at the mid-point of the socket length. The diameter of the socket may be decreased from the mouth to the root; for all pipe sizes, the total included angle of taper shall not exceed 0° 30'.

2 Only the manufacturer of the pipe is equipped to measure the socket inside diameter. Since the socket length is minimum (no tolerance is given to this dimension), it is not practical, other than for the manufacturer, to establish the exact position of the midpoint of the socket. He can therefore, tool up to measure his own pipe but such equipment will not necessarily give the correct figures for a pipe of other manufacturer.

Table 3 Dimensions of Sockets for Solvent Cement Jointing

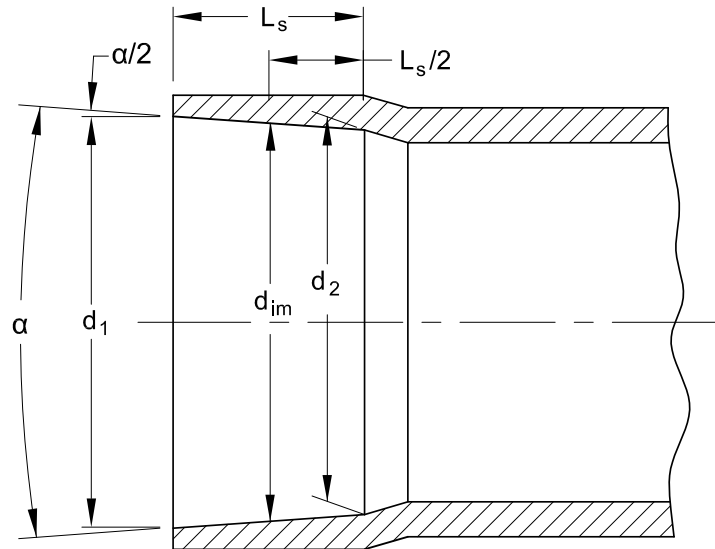
(Clause 7.2.1.1)

(Based on ISO 727)

All dimensions in millimetres.

Sl No.	Nominal Size DN	Socket Length L_s	Mean Socket Internal Diameter at Mid-Point of Socket Length, d_{im}	
			Min	Max
(1)	(2)	(3)	(4)	(5)
i)	20	16.0	20.1	20.3
ii)	25	19.0	25.1	25.3
iii)	32	22.0	32.1	32.3
iv)	40	26.0	40.1	40.3
v)	50	31.0	50.1	50.3
vi)	63	37.5	63.1	63.3
vii)	75	43.5	75.1	75.3
viii)	90	51.0	90.1	90.3
ix)	110	61.0	110.1	110.4
x)	125	68.5	125.1	125.4
xi)	140	76.0	140.2	140.5
xii)	160	86.0	160.2	160.5
xiii)	180	96.0	180.2	180.5
xiv)	200	106.0	200.3	200.6
xv)	225	118.5	225.3	225.7
xvi)	250	131.0	250.4	250.8
xvii)	280	146.0	280.4	280.9
xviii)	315	163.5	315.4	316.0
xix)	355	183.5	355.4	356.0
xx)	400	206.0	400.4	401.0
xxi)	450	231.0	450.4	451.0
xxii)	500	256.0	500.4	501.0
xxiii)	560	286.0	560.4	561.0
xxiv)	630	321.0	630.4	631.0

NOTE — For nominal sizes 20 mm to 225 mm, the dimensions are based on ISO 727-1985 (E).



- d_{im} = DIAMETER AT MID POINT OF SOCKET LENGTH
 d_1 = DIAMETER AT MOUTH
 d_2 = DIAMETER AT ROOT
 L_s = SOCKET LENGTH
 α = TOTAL INCLUDED ANGLE OF TAPER

FIG. 3 SOCKET DIMENSIONS FOR SOLVENT CEMENT JOINTS

7.2.1.2 Sockets for elastomeric sealing ring joints

These shall conform to the dimensions given in Table 4 read along with Fig. 4. The wall thickness of the elastomeric sealing ring type socket at any point except at sealing ring groove, shall not be less than the minimum specified wall thickness of the connecting pipes, and the same at the sealing ring groove shall not be less than 0.8 times the minimum specified wall thickness of the connecting pipe.

- a) Minimum depth of engagement for sockets for use with elastomeric sealing rings is calculated from the following formulae and rounded off to the next higher integer and shall conform to Table 5 read along with Fig. 5:

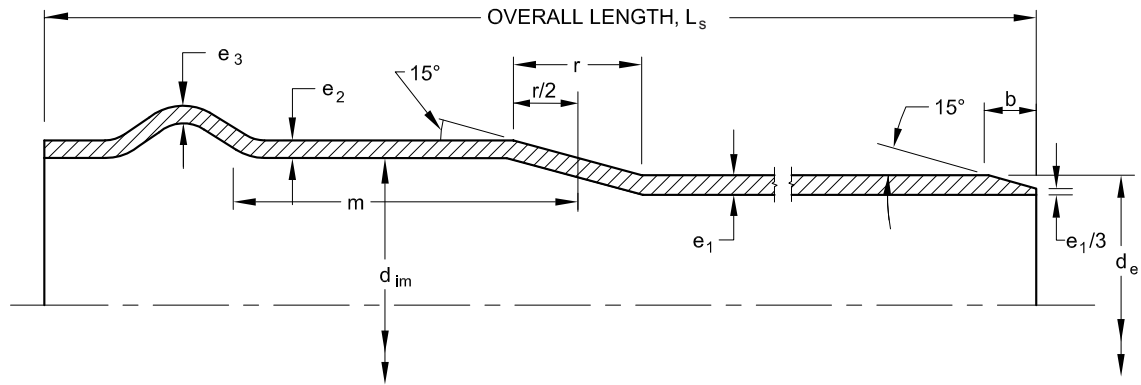
- 1) For nominal diameters $d_n < 280$ mm : $m > 50 \text{ mm} + 0.22 d_n$, and
- 2) For nominal diameters $d_n \geq 280$ mm : $m > 70 \text{ mm} + 0.15 d_n$.

where

m = minimum depth of engagement.

- b) Maximum inner diameter of groove in combination with the inner diameter of the sealing ring and the average outer diameter of the pipe shall ensure that the joint conforms to the required pressure rating as given in Table 7.

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m = MINIMUM DEPTH OF ENGAGEMENT
 d_e = OUTER DIAMETER AT ANY POINT
 d_{im} = INNER DIAMETER OF SOCKET, MEASURED AT MID POINT OF SOCKET LENGTH ' m '

FIG. 4 SOCKET FOR USE WITH ELASTOMERIC SEALING RINGS

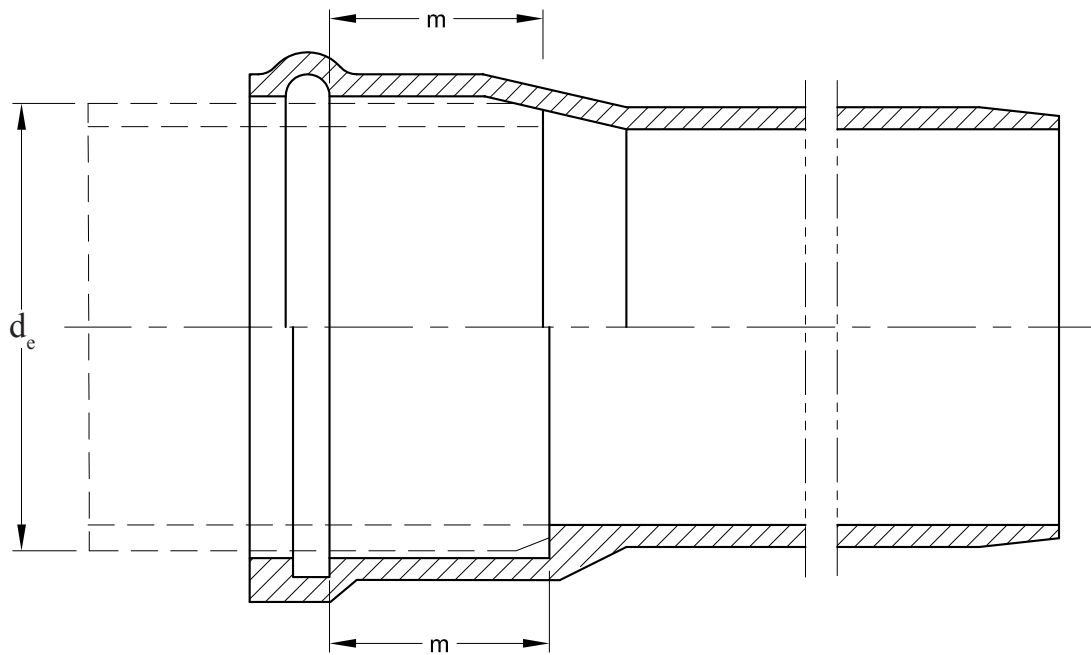


FIG. 5 MINIMUM DEPTH OF ENGAGEMENT

Table 4 Inner Diameters of Sockets for Use with Elastomeric Sealing Rings

(Clause 7.2.1.2)

All dimensions in millimetres.

SI No.	Nominal Outer Diameter	Socket Inner Diameter
(1)	d_n (2)	d_{im} (3)
i)	63	$63.6^{+0.7}_{-0.0}$
ii)	75	$75.6^{+0.7}_{-0.0}$
iii)	90	$90.7^{+0.8}_{-0.0}$
iv)	110	$110.8^{+0.9}_{-0.0}$
v)	125	$125.9^{+1.0}_{-0.0}$
vi)	140	$140.9^{+1.0}_{-0.0}$
vii)	160	$161.0^{+1.1}_{-0.0}$
viii)	180	$181.1^{+1.1}_{-0.0}$
ix)	200	$201.2^{+1.1}_{-0.0}$
x)	225	$226.4^{+1.1}_{-0.0}$
xi)	250	$251.5^{+1.1}_{-0.0}$
xii)	280	$281.6^{+1.4}_{-0.0}$
xiii)	315	$316.8^{+1.5}_{-0.0}$
xiv)	355	$357.0^{+1.7}_{-0.0}$
xv)	400	$402.2^{+1.9}_{-0.0}$
xvi)	450	$452.5^{+2.0}_{-0.0}$
xvii)	500	$502.6^{+2.0}_{-0.0}$
xviii)	560	$562.8^{+2.4}_{-0.0}$
xix)	630	$633.1^{+2.6}_{-0.0}$

Table 5 Minimum Depth of Engagement

[Clause 7.2.1.2 (a)]

All dimensions in millimetres.

SI No.	Nominal Outer Diameter of Pipe	Minimum Depth of Engagement
(1)	d_n (2)	m (3)
i)	63	64
ii)	75	67
iii)	90	70
iv)	110	75
v)	125	78
vi)	140	81
vii)	160	86
viii)	180	90
ix)	200	94
x)	225	100
xi)	250	105
xii)	280	112
xiii)	315	118
xiv)	355	124
xv)	400	130
xvi)	450	138
xvii)	500	145
xviii)	560	154
xix)	630	165

8 SEALING RINGS

Sealing rings shall be of any of the six hardness clauses as per IS 5382. The manufacturer has to however specify the hardness class and application type of sealing ring that is being offered. The design of the profile of the sealing ring is left to the manufacturer as long as the pipe with sealing ring meets the requirements of the standard.

NOTE — A test report or conformity certificate may be obtained from the manufacturer of the sealing ring for conformity to IS 5382. The frequency of this test report or conformity certificate shall be once in three months.

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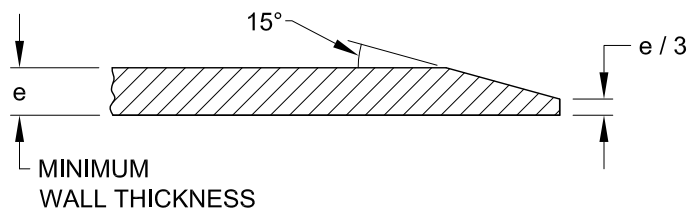


FIG. 6 PIPE ENDS

9 PIPE ENDS

9.1 The ends of the pipes meant for solvent cementing (both plain and bell ended) shall be cleanly cut and shall be reasonably square to the axis of the pipe or may be chamfered at the plain end.

9.2 Pipes with plain end(s) to be used for elastomeric sealing ring type joints shall be chamfered at approximately 15° to the axis of the pipe. Approximately two thirds of the full wall thickness shall be chamfered as shown in Fig. 6.

10 PHYSICAL AND CHEMICAL CHARACTERISTICS

10.1 Visual Appearance

The colour of the pipes shall be light grey (*see also 13.1.2*). Slight variations in the appearance of the colour are permitted. The pipes may also be supplied in any other colour as agreed to between the buyer and the seller.

10.1.1 The internal and external surfaces of the pipe shall be smooth, clean and free from grooving and other defects. Slight shallow longitudinal grooves or irregularities in the pipe shall be permissible provided the wall thickness remains within the permissible limits.

10.2 Opacity

The wall of the plain pipe shall not transmit more than 0.2 percent of the visible light falling on it when tested in accordance with IS 12235 (Part 3). The convex (outer) surface of the pipe specimen shall face the light source.

10.3 Effect on Water

The pipes shall not have any detrimental effect on the composition of water flowing through them. When tested by the method described in IS 12235 (Part 4) and IS 12235 (Part 10), the quantities of lead, dialkyl tin C4 and higher homologues (measured as tin), and any other toxic substances extracted from the internal

walls of the pipes shall not exceed the following concentrations in the test solution:

Lead (first extraction)	:	1.0 mg/l (1.0 ppm by mass)
Lead (third extraction)	:	0.05 mg/l (0.05 ppm by mass)
Dialkyl tin C4 and higher homologues measured as tin (third extraction)	:	0.02 mg/l (0.02 ppm by mass)
Cadmium (for all three extracts)	:	0.01 mg/l (0.01 ppm by mass)
Mercury (for all three extracts)	:	0.001 mg/l (0.001 ppm by mass)
Other toxic substances such as 'di-n-octyl-tin- mass) s-s bis iso-octyl mercapto acetate' and 'butyl stearate' (third extraction)	:	0.01 mg/l (0.01 ppm by mass)

10.3.1 The manufacturer, for the purpose of these tests, shall disclose any other toxic substances present. The limit of all the 'other toxic substances', including the two mentioned in **10.3** shall not exceed the specified limit of 0.01 mg/l.

NOTE — Implementation of the phase out programme for use of lead stabilizers in PVC pipe and fitting manufacturing of Government of India shall be borne in mind.

10.4 Reversion Test

When tested by the immersion method prescribed in IS 12235 (Part 5/Sec 1) and (Part 5/Sec 2), a length of pipe 200 ± 20 mm long shall not alter in length by more than 5 percent. In the case of socket end pipes, this test shall be carried out on the plain portion of the pipe taken at least 100 mm away from the root of the socket.

10.5 Vicat Softening Temperature

When tested by the method prescribed in IS 12235 (Part 2), the Vicat Softening temperature of the specimen shall not be less than 80°C .

10.6 Density

When determined in accordance with IS 12235 (Part 14), the density of the pipe shall be between 1.40 and 1.46 g/cm³.

10.7 Sulphated Ash Content Test

When tested as per Annex B, the sulphated ash content in the pipe shall not exceed 11 percent.

11 MECHANICAL PROPERTIES

11.1 Hydrostatic Characteristics

When subjected to internal hydrostatic pressure test in accordance with the procedure given in IS 12235 (Part 8/Sec 1), the pipe shall not fail during the prescribed test duration. The temperatures, duration and test pressures of the test shall conform to the requirements given in Table 6. The test shall be carried out not earlier than 24 h after the pipes have been manufactured. When tested in accordance with the method prescribed in IS 12235 (Part 8/Sec 4), the joints made with elastomeric sealing ring sockets shall fulfil the requirements given in Table 7. The requirements for integral sealing ring sockets are given in Table 7.

11.1.1 Acceptance test at 27 °C as given in Table 6 shall not apply to plumbing pipes. For plumbing pipes the test pressure for acceptance test at 27 °C shall be 3.6 MPa for 1 h, as these pipes are designed with a higher wall thickness for rigidity and not for providing a higher working pressure.

The type tests do not apply to plumbing pipes due to the same reason as above.

11.2 Resistance to External Blows at 0 °C

When tested by the method prescribed in Annex C, the pipe shall have a true impact rate of not more than 10 percent. In case of socket-ended pipes, this test shall be carried out on the plain portion of the pipe taken at least 100 mm away from the root of the socket.

12 SAMPLING AND CRITERIA FOR CONFORMITY

The sampling procedure and the criteria for conformity shall be as given in Annex D.

13 MARKING

13.1 Each pipe shall be clearly and indelibly marked in colour using ink/paint as per **13.1.1** at intervals of not more than 3 m. Alternatively, inkjet printing in any contrasting colour can also be used for marking at intervals of not more than 3 m. The markings shall show the following:

- Manufacturer's name or trade-mark,
- Outside diameter,
- Class of pipe and pressure rating,
- Batch or lot number, and
- The word plumbing in the case of plumbing pipes.

13.1.1 The information according to **13.1** shall be marked in colour as indicated below for different classes of pipes (in the case of indelible marking by ink/paint). In case of inkjet printing, the pipes shall also be provided near the end with a circumferential colour band as indicated below for different classes of pipes to identify the class of pipe:

Class of Pipe	Colour
Class 1	Red
Class 2	Blue
Class 3	Green
Class 4	Brown
Class 5	Yellow
Class 6	Black
Plumbing pipes	Pink

13.1.2 The pipes for water supply, Type A shall additionally bear continuous longitudinal blue colour strip printing and these longitudinal blue colour strip shall be so placed so as to not merge/disturb the information marked on the pipes as per **13.1**.

13.1.3 The lot number/batch number shall include the details of production in the following manner:

Year	Month	Day	Machine No.	Shift
XXXX	XX	XX	XXX	X

Table 6 Requirements of Pipes for Internal Hydrostatic Pressure Test

(Clauses 11.1 and 11.1.1)

Sl No.	Test	Test Temperature, <i>Min</i> °C	Test Duration (Minimum Holding Time) h	Test Pressure, <i>Min</i> MPa
(1)	(2)	(3)	(4)	(5)
i)	Type test	60	1 000	$1.16 \times PN$ (MPa)
ii)	Acceptance test	27	1	$4.19 \times PN$ (MPa)

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13.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity

assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

Table 7 Requirements of Integral Sealing Ring Sockets for Internal Hydrostatic Pressure Test

[*Clauses 7.2.1.2 (b) and 11.1*]

Sl No.	Diameter Range	Test	Test Temperature (Min)	Time	Test Pressure (Min)
	Mm		°C	h	MPa
(1)	(2)	(3)	(4)	(5)	(6)
i)	$d_n < 90$	Acceptance test	27	1	$2.88 \times PN$ (MPa)
		Type test	27	1 000	$2.20 \times PN$ (MPa)
ii)	$d_n \geq 90$	Acceptance test	27	1	$3.60 \times PN$ (MPa)
		Type test	27	1 000	$2.74 \times PN$ (MPa)

ANNEX A

(Clause 2)

LIST OF REFERRED INDIAN STANDARDS

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
4669 : 1968	Method of test for polyvinyl chloride resins	(Part 3) : 2004	Test for opacity
4905 : 2015/ ISO 24153 : 2009	Random sampling and randomization procedures (<i>first revision</i>)	(Part 4) : 2004	Determining the detrimental effect on the composition of water
5382 : 2018/ ISO 4633 : 2015	Rubber seals — Joint rings for water supply, drainage and sewerage pipelines — Specification for materials (<i>second revision</i>)	(Part 5/Sec 1) : 2004	Longitudinal reversion, Section 1 Determination methods
10148 : 1982	Positive list of constituents of polyvinyl chloride (PVC) and its copolymers in contact with foodstuffs, pharmaceuticals and drinking water	(Part 5/Sec 2) : 2004	Longitudinal reversion, Section 2 Determination parameters
10151 : 2019	Polyvinyl chloride (PVC) and its copolymers for its safe use in contact with foodstuffs, pharmaceuticals and drinking water — Specification (<i>first revision</i>)	(Part 8/Sec 1) : 2004	Resistance to internal hydrostatic pressure, Section 1 Resistance to internal hydrostatic pressure at constant internal water pressure
12231 : 1987	Specification for unplasticized PVC pipes for use in suction and delivery lines of agricultural pump sets	(Part 8/Sec 4) : 2004	Resistance to internal hydrostatic pressure, Section 4 Leak-tightness of elastomeric sealing ring type socket joints under positive internal pressure without angular deflection
12235	Thermoplastics pipes and fittings — Methods of test (<i>first revision</i>)	(Part 10) : 2004	Determination of organotin as tin aqueous solution
(Part 1) : 2004	Measurement of dimensions	(Part 14) : 2004	Determination of density/relative density (specific gravity)
(Part 2) : 2004	Determination of Vicat softening temperature	12818 : 2010	Unplasticized polyvinyl chloride (PVC-U) screen and casing pipes for bore/tubewells — Specification (<i>second revision</i>)

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ANNEX B

(Clause 10.7)

SULPHATED ASH CONTENT TEST

B-1 PRINCIPLE

Calcination with sulphuric acid treatment after combustion, that is, by burning the substance and transforming the residue into sulphates using concentrated sulphuric acid and, finally, heating the residue at 850 °C until constant mass is reached.

B-2 REAGENTS

B-2.1 Sulphuric Acid, density 1 840 kg/m³.

B-3 APPARATUS

B-3.1 Silica or platinum crucible, inert to the material tested. The size shall be sufficient so that the crucible is not more than half filled by the test portion sample.

B-3.2 Analytical Balance, with 0.1 mg accuracy.

B-3.3 Bunsen Burner, with silica triangle and tripod or other suitable heating device.

B-3.4 Muffle Furnace, capable of being maintained at 850 ± 10 °C.

B-3.5 Pipette, of appropriate capacity.

B-3.6 Dessicator containing an effective drying agent that does not react chemically with the ash components.

NOTE — In some cases, the affinity of the ash for water may be greater than that of drying agents commonly used.

B-4 PROCEDURE

B-4.1 Prepare the crucible by heating in the muffle furnace at 850 ± 10 °C until constant mass is reached. Allow it to cool in the dessicator to room temperature, but for at least one hour, and weigh to the nearest 0.1 mg (M_1).

B-4.2 Introduce into the crucible 2 g to 5 g of the sample and reweigh to the nearest 0.1 mg (M_2). Heat the crucible directly on the heating device so that the sample burns slowly and loss of ash is avoided. Continue this operation until no more smoke is evolved.

B-4.3 After allowing the crucible and contents to cool, add sulphuric acid drop wise by means of a pipette of suitable capacity until the residue is soaked completely. Heat carefully on the heating device until the evolution

of smoke ceases, taking care to avoid spattering of the contents of the crucible.

B-4.4 If, after allowing the crucible to cool, carbon is still evident, add 1 to 5 drops of sulphuric acid and reheat until evolution of white fumes has ceased.

B-4.5 Place the crucible at the entrance of the muffle furnace maintained at 850 ± 10 °C (the temperature in the entrance zone is about 300 to 400 °C), then advance the crucible slowly into the furnace. Calcine slowly (to prevent loss of ash particles) for 30 min at 850 ± 10 °C.

B-4.6 Remove the crucible from the furnace. Place it in the dessicator, allow to cool to room temperature, but for at least 1 h, and weigh to the nearest 0.1 mg (M_3).

B-4.7 Calcine again, under the same conditions until constant mass is reached, that is, until the results of two consecutive weighings do not differ by more than 0.5 mg. The duration of heating in the furnace shall not, however, exceed 3 h if constant mass is not attained after this time, the mass after 3 h shall be used for calculating the test result. The residue after calcination shall be white.

B-5 NUMBER OF DETERMINATIONS

Carry out two determinations. Calculate the arithmetic mean of the results. If the individual test results differ from each other by more than 10 percent of their mean, repeat the procedure until two successive results do not differ from each other by more than 10 percent of their mean.

B-6 EXPRESSION OF RESULTS

The sulphated ash content shall be calculated as follows:

Sulphated ash content, percent by mass =

$$\frac{M_3 - M_1}{M_2 - M_1} \times 100$$

where

M_1 = mass of the crucible,

M_2 = mass of the crucible and test portion sample,
and

M_3 = mass of the crucible and residue.

ANNEX C

(Clause 11.2)

METHOD OF TEST FOR RESISTANCE TO EXTERNAL BLOWS AT 0 °C

C-1 GENERAL

This annex specifies the method for the determination of the resistance to external blows of UPVC pipes manufactured according to this standard.

C-2 DEFINITIONS

For the purpose of this annex, the following definitions shall apply.

C-2.1 True Impact Rate (TIR) — The total number of failures divided by the total number of blows, as a percentage, as if the whole batch had been tested.

NOTE — In practice, test pieces are drawn at random from the batch and the result is only an estimate of the TIR for that batch.

C-2.2 Failure — Shattering or any crack or split on the inside of the pipe that was caused by the impact and that can be seen by the naked eye (lighting devices may be used to assist in examining the specimens).

C-2.2.1 Indentation of the test piece is not considered a failure.

C-3 PRINCIPLE

C-3.1 Test pieces are subjected to blows from a falling striker, of specified mass and shape, dropped from a known height on to specified positions around the circumference of the test piece. The true impact rate (TIR) of the batch, or production run from an extruder, is estimated.

C-3.2 The severity of this test method can be adjusted by changing the mass of the striker and/or by changing the drop height. It is not technically correct to vary the severity of the test by choosing values of the TIR other than those specified below.

The maximum acceptable values for the TIR is taken to be 10 percent.

NOTE — It shall be appreciated that a completely definitive result can be reached only by testing the whole batch, but in practice, a balance is necessary between the statistical possibility of a definitive result and the cost of further testing.

C-4 APPARATUS

C-4.1 Falling Weight Testing Machine

It shall incorporate the basic components as given in C-4.1.1 to C-4.1.4 (see Fig. 7).

C-4.1.1 Main Frame, with guide rails or tube, which can be fixed in the true vertical position, to accommodate a striker (see C-4.1.2) and release mechanism to release the striker to fall vertically and freely.

C-4.1.2 Striker, having a nose comprising all or part of a hemisphere, combined with a stem at least 10 mm long, and having dimensions conforming to Table 8 read along with Fig. 8. The mass of the striker, including any associated weights, shall be selected from the values given in Table 9. Below the stem, the nose shall be of solid steel, polished and free from flats, indentations or other imperfections which may influence the result.

C-4.1.3 Rigid Specimen Support, consisting of a 120° V-block at least 200 mm long, positioned so that the vertical projection of the point of impact of the falling striker is within 2.5 mm of the axis of the V-block (see Fig. 7).

Table 8 Dimensions for the Nose of the Striker

(Clause C-4.1.2)

All dimensions in millimetres.

Sl No.	Type	R_s	d	d_s	α
(1)	(2)	(3)	(4)	(5)	(6)
i)	d_{25}	50	25 ± 1	Free	Free
ii)	d_{90}	50	90 ± 1	Free	Free

Table 9 Classified Striker Mass and Drop Height Conditions for the Falling Weight Impact Test

(Clauses C-4.1.2 and C-8.1)

Sl No.	Nominal Outside Diameter of Pipes, d_n mm	Mass of Falling Weight kg	Fall Height mm
(1)	(2)	(3)	(4)
i)	Up to and including 25	$0.25 \pm 0.5 \%$	500 ± 10
ii)	32	$0.25 \pm 0.5 \%$	$1\ 000 \pm 10$
iii)	40	$0.25 \pm 0.5 \%$	$1\ 000 \pm 10$
iv)	50	$0.25 \pm 0.5 \%$	$1\ 000 \pm 10$
v)	63	$0.25 \pm 0.5 \%$	$2\ 000 \pm 10$
vi)	75	$0.25 \pm 0.5 \%$	$2\ 000 \pm 10$
vii)	90	$0.50 \pm 0.5 \%$	$2\ 000 \pm 10$
viii)	110	$0.50 \pm 0.5 \%$	$2\ 000 \pm 10$
ix)	125 and above	$1.00 \pm 0.5 \%$	$2\ 000 \pm 10$

C-4.1.4 Release Mechanism, such that the striker can fall from a variable height which can be adjusted to any height up to at least 2 m, measured from the top surface of the test piece, with an accuracy of ± 10 mm.

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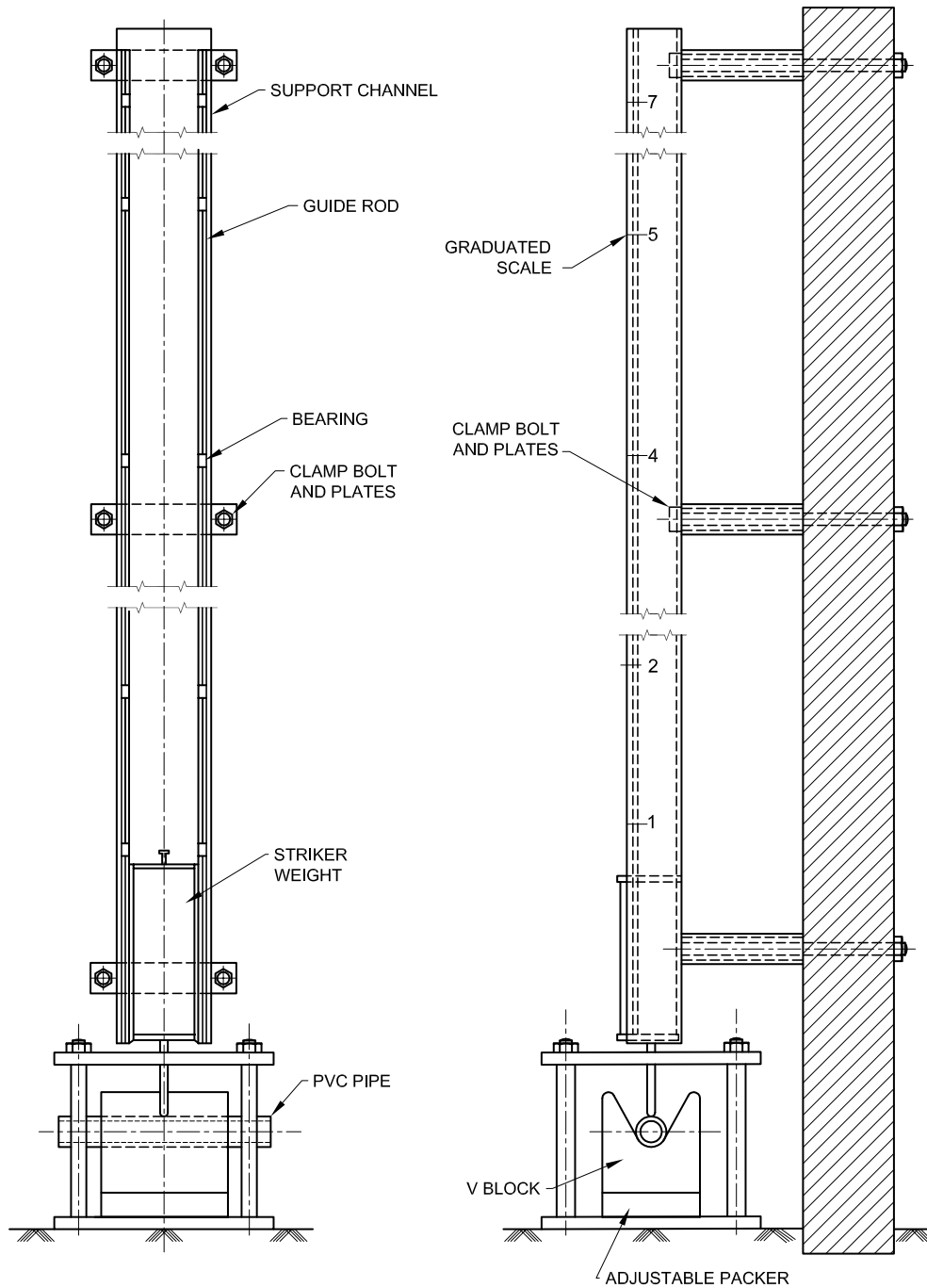
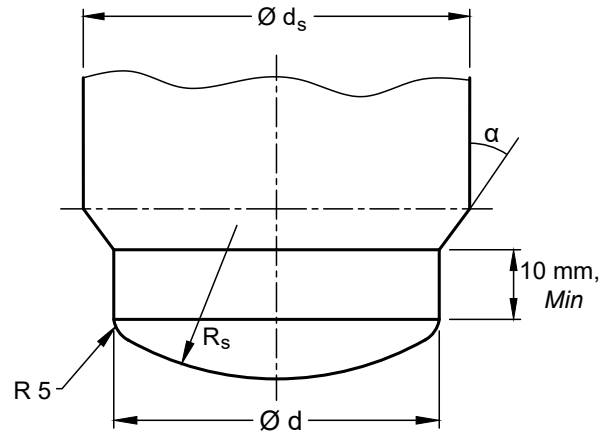
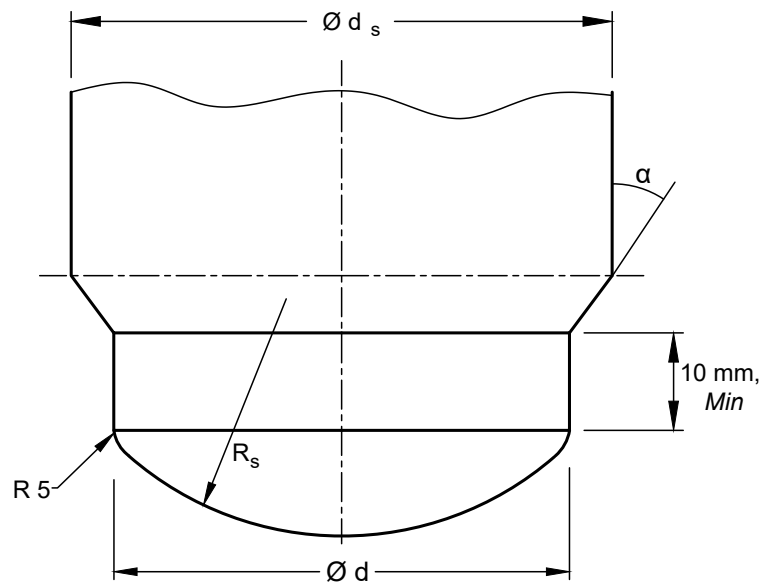


FIG. 7 IMPACT TESTING MACHINE



8A TYPE d_{25} (FOR STRIKERS OF MASS 0.25 kg AND 0.5 kg)



8B TYPE d_{90} (FOR STRIKERS OF MASS EQUAL TO OR GREATER THAN 1 kg)

FIG. 8 NOSES OF THE STRIKERS

C-5 TEST PIECES

Test pieces of length 200 ± 10 mm shall be cut from the pipe selected at random from the batch, or the production run from an extruder. The cut ends shall be square to the axis of the pipe, clean and free from damage. For pipes with outside diameters greater than

40 mm, a straight line shall be drawn along the length of each test piece at a random position. Further lines shall be drawn at equal distances around the pipe piece so that each test piece has a number of lines given in Table 10. The number of blows required is given in C-6. For pipes with outside diameters less than or equal to 40 mm, only one blow per test piece shall be made.

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Table 10 Number of Equidistant Lines to be drawn on Test Pieces
(Clause C-5)

Sl No.	Nominal Outside Diameter of Pipes mm	Number of Equidistant Lines to be Drawn mm
(1)	(2)	(3)
i)	Up to and including 40	–
ii)	50	3
iii)	63	3
iv)	75	4
v)	90	4
vi)	110	6
vii)	125	6
viii)	140	8
ix)	160	8
x)	180	8
xi)	200	12
xii)	225	12
xiii)	250	12
xiv)	280 and above	16

C-6 SAMPLING TO CONFIRM VALUE OF TIR ON ISOLATED BATCHES

C-6.1 If the number of failures from a sample falls into Region A of Fig. 9 (for a TIR of less than or equal to 10 percent), then reasonable confirmation is obtained that the batch has a TIR less than or equal to the specified level.

C-6.2 If the number of failures falls into the Region C of Fig. 9, the batch can be judged to have a TIR greater than the specified value.

C-6.3 If the number of failures falls into the Region B of Fig. 9, in general further test pieces should be taken so that a decision shall be arrived at.

C-6.4 The decision shall be made by using the cumulative result of all the test pieces examined from the batch under consideration.

C-6.5 Fig. 9 is a guideline to indicate the principle of the test method. Evaluation of the test result shall be obtained based on Table 11. If the number of blows exceed 124, Fig. 9 shall be referred for assessment of the result.

C-7 CONDITIONING

C-7.1 The test pieces shall be conditioned in a liquid bath or in air at a temperature of 0 ± 1 °C for at least the period given in Table 12.

C-7.1.1 In case of disputes over the results, a liquid bath shall be used.

C-7.2 Test pieces with wall thickness up to 8.6 mm shall be tested within 10 s of their removal from air conditioning, or within 20 s of their removal from liquid conditioning, as applicable.

C-7.3 Test pieces with wall thickness greater than 8.6 mm shall be tested within 20 s of their removal from air conditioning or within 30 s of their removal from liquid conditioning, as applicable.

C-7.4 If this interval is exceeded, the test piece shall be returned immediately to the unit for reconditioning for further period of at least 10 min.

Boundaries between regions are calculated using the following equations:

$$S_{A/B} = np - 0.5 \mu \sqrt{np(1-p)}$$

$$S_{B/C} = np + 0.5 \mu \sqrt{np(1-p)}$$

where

$\mu = 1.282$ (10 percent one-sided)

$p = 0.10$ (TIR), and

$n =$ number of blows.

NOTES

1 Initially, a minimum of 25 blows shall be made. In case of no failure, the lot is deemed to have passed the test. In case of four or more failures, the lot is treated as rejected. The test shall be continued further if one or two or three failures occur till the results fall into either Region A or Region C of Table 11 to arrive at a decision for acceptance or rejection, respectively.

2 It is necessary to have achieved at least 25 blows without failure before the test is discontinued.

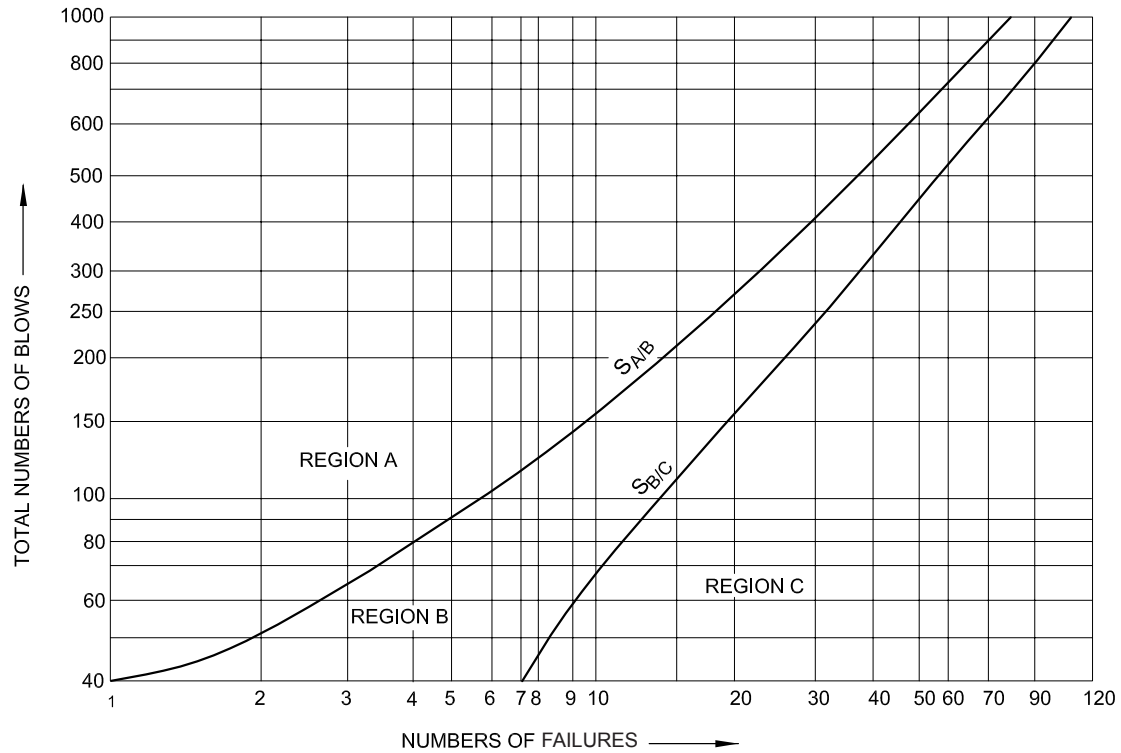


FIG. 9 NUMBER OF TEST PIECES FOR 10 PERCENT TIR
(AT 90 PERCENT CONFIDENCE LEVEL)

Table 11 Number of Blows and Failures
(Clause C-6.5)

Sl No.	Number of Blows	Number of Failures			Sl No.	Number of Blows	Number of Failures		
		Accept Region A	Continue Test Region B	Reject Region C			Accept Region A	Continue Test Region B	Reject Region C
(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
i)	25	0	1 to 3	4	xvi)	40	1	2 to 6	7
ii)	26	0	1 to 4	5	xvii)	41	1	2 to 6	7
iii)	27	0	1 to 4	5	xviii)	42	1	2 to 6	7
iv)	28	0	1 to 4	5	xix)	43	1	2 to 6	7
v)	29	0	1 to 4	5	xx)	44	1	2 to 6	7
vi)	30	0	1 to 4	5	xxi)	45	1	2 to 6	7
vii)	31	0	1 to 4	5	xxii)	46	1	2 to 6	7
viii)	32	0	1 to 4	5	xxiii)	47	1	2 to 6	7
ix)	33	0	1 to 5	6	xxiv)	48	1	2 to 6	7
x)	34	0	1 to 5	6	xxv)	49	1	2 to 7	8
xi)	35	0	1 to 5	6	xxvi)	50	1	2 to 7	8
xii)	36	0	1 to 5	6	xxvii)	51	1	2 to 7	8
xiii)	37	0	1 to 5	6	xxviii)	52	1	2 to 7	8
xiv)	38	0	1 to 5	6	xxix)	53	2	3 to 7	8
xv)	39	0	1 to 5	6	xxx)	54	2	3 to 7	8

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Table 11 (Concluded)

SI No.	Number of Blows	Number of Failures			SI No.	Number of Blows	Number of Failures		
		Accept Region A	Continue Test Region B	Reject Region C			Accept Region A	Continue Test Region B	Reject Region C
(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
xxxvi)	55	2	3 to 7	8	lxxxi)	95	5	6 to 12	13
xxxvii)	56	2	3 to 7	8	lxxxii)	96	5	6 to 12	13
xxxviii)	57	2	3 to 8	9	lxxxiii)	97	5	6 to 12	13
xxxix)	58	2	3 to 8	9	lxxxiv)	98	5	6 to 13	14
xl)	59	2	3 to 8	9	lxxxv)	99	5	6 to 13	14
xli)	60	2	3 to 8	9	lxxxvi)	100	5	6 to 13	14
xlii)	61	2	3 to 8	9	lxxxvii)	101	5	6 to 13	14
xliii)	62	2	3 to 8	9	lxxxviii)	102	5	6 to 13	14
xliv)	63	2	3 to 8	9	lxxxix)	103	5	6 to 13	14
xlvi)	64	2	3 to 8	9	lxxx)	104	5	6 to 13	14
xlvi)	65	2	3 to 9	10	lxxxii)	105	6	7 to 13	14
xlvi)	66	2	3 to 9	10	lxxxiii)	106	6	7 to 14	15
xlvi)	67	3	4 to 9	10	lxxxiv)	107	6	7 to 14	15
xlvi)	68	3	4 to 9	10	lxxxv)	108	6	7 to 14	15
xlvi)	69	3	4 to 9	10	lxxxvi)	109	6	7 to 14	15
xlvi)	70	3	4 to 9	10	lxxxvii)	110	6	7 to 14	15
xlvi)	71	3	4 to 9	10	lxxxviii)	111	6	7 to 14	15
xlvi)	72	3	4 to 9	10	lxxxix)	112	6	7 to 14	15
xlvi)	73	3	4 to 10	11	xc)	113	6	7 to 14	15
xlvi)	74	3	4 to 10	11	xc)	114	6	7 to 15	16
xlvi)	75	3	4 to 10	11	xc)	115	6	7 to 15	16
xlvi)	76	3	4 to 10	11	xc)	116	6	7 to 15	16
xlvi)	77	3	4 to 10	11	xc)	117	7	8 to 15	16
xlvi)	78	3	4 to 10	11	xc)	118	7	8 to 15	16
xlvi)	79	3	4 to 10	11	xc)	119	7	8 to 15	16
xlvi)	80	4	5 to 10	11	xc)	120	7	8 to 15	16
xlvi)	81	4	5 to 11	12	xc)	121	7	8 to 15	16
xlvi)	82	4	5 to 11	12	xc)	122	7	8 to 15	16
xlvi)	83	4	5 to 11	12	xc)	123	7	8 to 16	17
xlvi)	84	4	5 to 11	12	c)	124	7	8 to 16	17
xlvi)	85	4	5 to 11	12	Table 12 Conditioning Period				
xlvi)	86	4	5 to 11	12	<i>(Clause C-7.1)</i>				
xlvi)	87	4	5 to 11	12					
xlvi)	88	4	5 to 11	12					
xlvi)	89	4	5 to 12	13					
xlvi)	90	4	5 to 12	13					
xlvi)	91	4	5 to 12	13					
xlvi)	92	5	6 to 12	13					
xlvi)	93	5	6 to 12	13					
xlvi)	94	5	6 to 12	13					

C-8 PROCEDURE

C-8.1 The mass of the falling striker and the drop height appropriate to the pipe shall be as specified in Table 9.

C-8.2 For pipes of outside diameter 40 mm or less, subject the test piece to a single blow only.

C-8.3 For pipes of outside diameter greater than 40 mm, subject the test piece to a blow by allowing the striker to fall on one of the marked lines. If the test piece passes

the test, rotate it in the V-block to the next marked line and again subject it to a blow from the falling striker, after reconditioning, if necessary (*see C-7*).

C-8.4 Continue this procedure until the test piece fails the test, or until all the marked lines have been struck one blow.

C-8.5 If required, carry out the test on subsequent test pieces, subjecting each one to the required number of blows.

ANNEX D

(Clause 12)

SAMPLING AND CRITERIA FOR CONFORMITY

D-1 ACCEPTANCE

D-1.1 Acceptance tests are carried out on samples selected from a lot for the purpose of acceptance of the lot.

D-1.2 Lot

All PVC pipes in a single consignment of the same class, same size and manufactured under essentially similar conditions shall constitute a lot.

D-1.3 For ascertaining conformity of the lot to the requirements of the standard, samples shall be tested from each lot separately.

D-1.4 Visual and Dimensional Requirements

D-1.4.1 The number of test samples to be taken from a lot shall depend on the size of the lot and the outside diameter of the pipes, and shall be in accordance with Table 13.

D-1.4.2 These pipes shall be selected at random from the lot and in order to ensure the randomness of selection, a random number table shall be used. For guidance and use of random number tables, IS 4905 may be referred to. In the absence of a random number table, the following procedure may be adopted:

Starting from any pipe in the lot, count them as 1, 2, 3, etc, up to r and so on, where r is the integral part of N/n , N being the number of pipes in the lot, and the number of pipes in the sample. Every r^{th} pipe so counted shall be withdrawn so as to constitute the required sample size.

D-1.4.3 The number of pipes given for the first sample in col 4 of Table 13, shall be taken from the lot and examined for visual and dimensional requirements given in 7 and 10.1. A pipe failing to satisfy any of these requirements shall be considered as defective. The lot shall be deemed to have satisfied these requirements, if the number of defectives found in the first sample is less than or equal to the corresponding acceptance

number given in col 6 of Table 13. The lot shall be deemed not to have met these requirements, if the number of defectives found in the first sample is greater than or equal to the corresponding rejection number given in col 7 of Table 13. If, however, the number of defectives found in the first sample lies between the corresponding acceptance and rejection numbers given in col 6 and 7, a second sample of the size given in col 4 shall be taken and examined for these requirements. The lot shall be considered to have satisfied these requirements if the cumulative sample is less than or equal to the corresponding acceptance number given in col 5, otherwise not.

D-1.5 Reversion Test

D-1.5.1 The lot, having satisfied visual and dimensional requirements, shall be tested for reversion.

D-1.5.2 For this purpose, the number of pipes given for the first sample in col 4 of Table 14 shall be taken from the lot. The sample pipe failing the reversion test shall be considered as defective. The lot shall be deemed to have met the requirements given in this specification for the reversion test, if the number of defectives found in the first sample is less than or equal to the corresponding acceptance number given in col 6. The lot shall be deemed not to have met these requirements, if the number of defectives found in the first sample is greater than or equal to the corresponding rejection number given in col 7. If, however, the number of defectives in the first sample lies between the corresponding acceptance and rejection numbers given in col 6 and col 7, a second sample of size given in col 4 shall be taken and examined for the requirement. The lot shall be considered to have satisfied the requirements, if the number of defectives found in the cumulative sample is less than or equal to the corresponding acceptance number given in col 6, otherwise not.

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Table 13 Scale of Sampling for Visual Appearance and Dimensional Requirements
(Clauses D-1.4.1 and D-1.4.3)

Sl No.	Number of Pipes in the Lots	Sample Number	Sample Size	Cumulative Sample Size	Acceptance Number	Rejection Number
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Up to 1 000	{ First	13	13	0	2
		{ Second	13	26	1	2
ii)	1 001 to 3 000	{ First	20	20	0	2
		{ Second	20	40	1	2
iii)	3 001 to 10 000	{ First	32	32	0	3
		{ Second	32	64	3	4
iv)	10 001 and above	{ First	50	50	1	4
		{ Second	50	100	4	5

Table 14 Scale of Sampling for Reversion, Vicat Softening Temperature and Density Test
(Clauses D-1.5.2, D-1.6.1 and D-1.7.2)

Sl No.	Number of Pipes in the Lot	Sample Number	Sample Size	Cumulative Sample Size	Acceptance Number	Rejection Number
(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>For d_n less than and up to 110 mm</i>						
i)	Up to 1 000	{ First	5	5	0	2
		{ Second	5	10	1	2
ii)	1 001 to 3 000	{ First	8	8	0	2
		{ Second	8	16	1	2
iii)	3 001 to 10 000	{ First	13	13	0	2
		{ Second	13	26	1	2
iv)	10 001 and above	{ First	20	20	0	3
		{ Second	20	40	3	4
<i>For d_n above 110 mm</i>						
v)	Up to 3 000	{ First	3	3	0	2
		{ Second	3	6	1	2
vi)	3 001 to 10 000	{ First	5	5	0	2
		{ Second	5	10	1	2
vii)	10 001 and above	{ First	8	8	0	2
		{ Second	8	16	1	2

D-1.6 Vicat Softening Test

D-1.6.1 The lot, having satisfied visual and dimensional requirements shall be tested for Vicat Softening temperature as given in Table 14.

D-1.6.2 For this purpose, the procedure adopted for sampling and criteria for conformity shall be the same as that for reversion under **D-1.5.2** using Table 14.

D-1.7 Density

D-1.7.1 The lot, having satisfied the visual and dimensional requirements, shall be tested for density.

D-1.7.2 For this purpose, the procedure adopted for sampling and criteria for conformity shall be the same as that for reversion under **D-1.5.2**, using Table 14.

D-1.8 Sulphated Ash Content Test

D-1.8.1 The lot, having satisfied the visual and dimensional requirements, shall be subjected to the sulphated ash content test.

D-1.8.2 For this purpose, the procedure adopted for sampling and criteria for conformity shall be as per Table 15.

Table 15 Scale of Sampling for Sulphated Ash Content Test
(Clause D-1.8.2)

SI No.	Number of Pipes in the Lot	Sample Number	Sample Size	Cumulative Sample Size	Acceptance Number	Rejection Number
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Up to 10 000	First	2	2	0	1
		Second	2	4	1	2
ii)	Above 10 000	First	3	3	0	2
		Second	3	6	1	2

Table 16 Scale of Sampling for Resistance to External Blows at 0 °C
(Clause D-1.9.2)

SI No.	Number of Pipes in the Lot	Sample Number	Sample Size	Cumulative Sample Size	Acceptance Number	Rejection Number
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Up to 3 000	First	3	3	0	2
		Second	3	6	1	2
ii)	3 001 to 10 000	First	5	5	0	2
		Second	5	10	1	2
iii)	10 001 and above	First	8	8	0	2
		Second	8	16	1	2

NOTE — The number mentioned in col 4 to col 7 represent the number of times the test is to be carried out and do not represent either the number of pipe samples or number of blows or number of failures.

D-1.9 Resistance to External Blows at 0 °C

D-1.9.1 The lot, having been found satisfactory according to **D-1.4**, **D-1.5**, **D-1.6**, **D-1.7** and **D-1.8**, shall be tested for resistance to external blows at 0 °C.

D-1.9.2 For this purpose, the procedure adopted for sampling and criteria for conformity shall be as specified in Annex C and Table 16.

D-1.10 Internal Hydrostatic Pressure Test (Acceptance Test)

D-1.10.1 The lot, having been found satisfactory according to **D-1.4**, **D-1.5**, **D-1.6**, **D-1.7**, **D-1.8** and **D-1.9**, shall be subjected to the requirements of the acceptance test for internal hydraulic pressure. The number of pipes to be taken from the lot shall depend on the size of the lot and shall be according to Table 17.

D-1.10.2 The pipes shall be taken at random from the lot. In order to ensure the randomness of selection, procedures given in IS 4905 may be followed.

D-1.10.3 Number of Tests and Criteria for Conformity

The number of test samples shall be as given in Table 17. The lot shall be considered to have satisfied the requirements for this test, if the number of test samples failing in this requirement is equal to the corresponding acceptance number given in col 4 of Table 17.

Table 17 Scale of Sampling for Internal Hydrostatic Test

(Clauses D-1.10.1 and D-1.10.3)

SI No.	Number of Pipes in the Lots	Sample Size	Acceptance Number
(1)	(2)	(3)	(4)
i)	Up to 3 000	2	0
ii)	3 001 to 10 000	3	0
iii)	10 001 and above	5	0

D-2 TYPE TESTS

D-2.1 Type tests are intended to prove the suitability and performance of a new composition or a new size of pipe. Such tests, therefore, need to be applied only when a change is made in polymer composition or when a new size of pipe is to be introduced. Type tests for compliance with **10.2**, **10.3** and **11.1** (type test only) shall be carried out.

D-2.1.1 Opacity

For this test, the manufacturer or the supplier shall furnish to the testing authority one sample of the pipe of the thinnest wall section, selected preferably from a regular production lot.

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D-2.1.1.1 The sample so selected shall be tested for compliance with requirements for opacity as given in **10.2**.

D-2.1.1.2 If the sample passes the requirements of the opacity test, the type of the pipe under consideration shall be considered to be eligible for approval, which shall be valid for a period of one year.

D-2.1.1.3 In case the sample fails in the test, the testing authority, at its discretion, may call for a fresh sample and subject the same to the opacity test. If the sample passes the repeat test, the type of pipe under consideration shall be considered eligible for approval. If the sample fails in the repeat test, the type of pipe shall not be approved. The manufacturer or the supplier may be asked to improve the design and resubmit the product for type approval.

D-2.1.1.4 At the end of the validity period (normally one year) or earlier, if necessary, the testing authority may call for a fresh sample for opacity test for the purpose of type approval.

D-2.1.2 *Test for Effect on Water*

For this type test, the manufacturer or the supplier shall furnish to the testing authority three samples of the smallest size of pipe taken from each machine (selected preferably from a regular production lot).

D-2.1.2.1 Three samples so selected shall be tested for compliance with the requirements for effect on water as given in **10.3**.

D-2.1.2.2 If all three samples pass the requirements for effect on water, the type test of the pipe under consideration shall be considered to be eligible for approval, which shall be normally valid for a period of one year.

D-2.1.2.3 In case any of the samples fails in this test, the testing authority, at its discretion, may call for fresh samples not exceeding the original number, and subject

them to the test for effect on water. If, in the repeat test, no single failure occurs, the type of pipe under consideration shall be considered eligible for type approval. If any of the samples fails in the repeat test, the type of pipe shall not be approved. The manufacturer or the supplier may be asked to improve the design and resubmit the product for type approval.

D-2.1.2.4 At the end of the validity period (normally one year) or earlier, if necessary, the testing authority may call for fresh samples for effect on water test for the purpose of type approval.

D-2.1.3 *Internal Hydrostatic Pressure Test (Type Test)*

For this type test, the manufacturer or the supplier shall furnish to the testing authority, three samples of pipes of different diameters and different classes (selected preferably from a regular production lot).

D-2.1.3.1 Three samples so selected shall be tested for compliance with the requirements of type test given in Table 6 and Table 7.

D-2.1.3.2 If all the three samples pass the requirements of the quality test, the type of pipe under consideration shall be considered to be eligible for type approval which shall be normally valid for a period of one year.

D-2.1.3.3 In case any of the samples fail in this test, the testing authority, at its discretion, may call for fresh samples not exceeding the original number and subject them to the type test. If, in the repeat test, no single failure occurs, the type of pipe shall be considered for type approval. If any of the samples fails in the repeat tests, the type of pipe shall not be approved. The manufacturer or the supplier may be asked to improve the design and resubmit the product for type approval.

D-2.1.3.4 At the end of the validity period (normally one year) or earlier, if necessary, the testing authority may call for fresh samples for type test for the purpose of type approval.

ANNEX E

(Foreword)

COMMITTEE COMPOSITION

Plastic Piping Systems Sectional Committee, CED 50

<i>Organization</i>	<i>Representative(s)</i>
Central Institute of Plastic Engineering and Technology, Chennai	DR S. K. NAYAK (Chairman)
Borouge India Pvt Ltd, Mumbai	SHRI PRASHANT D. NIKHADE
Brihan Mumbai Licensed Plumbers' Association, Mumbai	SHRI KISHOR V. MERCHANT SHRI BIJAL M. SHAH (<i>Alternate</i>)
Central Institute of Plastic Engineering and Technology, Chennai	DR S. N. YADAV SHRI D. ANJANEYA SHARMA (<i>Alternate</i>)
Central Public Health and Environmental Engineering Organization, New Delhi	ADVISER (PHE) ASSISTANT ADVISER (PHE) (<i>Alternate</i>)
Central Public Works Department, New Delhi	SHRI M. K. SHARMA SHRI AMAR SINGH (<i>Alternate</i>)
Chennai Metropolitan Water Supply & Sewerage Board, Chennai	ENGINEERING DIRECTOR SUPERINTENDING ENGINEER (P & D) (<i>Alternate</i>)
CSIR-Central Building Research Institute, Roorkee	DR B. SINGH SHRI RAJIV KUMAR (<i>Alternate</i>)
CSIR-National Environmental Engineering Research Institute, Nagpur	DR (SHRIMATI) ABHA SARGONKAR DR RITESH VIJAY (<i>Alternate</i>)
Delhi Development Authority, New Delhi	SUPERINTENDING ENGINEER (D) EXECUTIVE ENGINEER (R & D) (<i>Alternate</i>)
Delhi Jal Board, New Delhi	SHRI Y. K. SHARMA SHRI S. L. MEENA (<i>Alternate</i>)
Department of Chemical & Petrochemicals, Govt of India, New Delhi	JOINT INDUSTRIAL ADVISOR
Finolex Industries Limited, Pune	SHRI ARUN SONAWANE SHRI D. J. SALUNKE (<i>Alternate</i>)
GAIL India Limited, New Delhi	SHRI MANISH KHANDELWAL SHRI KULDEEP NEGI (<i>Alternate I</i>) SHRI NITIN GUPTA (<i>Alternate II</i>)
Government e-Marketplace, New Delhi	REPRESENTATIVE
Haldia Petrochemicals Ltd, Kolkata	SHRI RAJ K. DATTA SHRI AMARTYA MAITY (<i>Alternate</i>)
HPCL – Mittal Energy Ltd, Noida	SHRI VINEET KUMAR GUPTA SHRI ALAKESH GHOSH (<i>Alternate</i>)
HSIL Ltd (Pipe Divison), Hyderabad	SHRI TUSHAR LOKARE SHRI VINOY KUMAR (<i>Alternate</i>)
Indian Oil Corporation Ltd, Panipat	SHRI DHANANJAY SAHOO SHRI RAJA PODDAR (<i>Alternate</i>)
Jain Irrigation System Limited, Jalgaon	SHRI S. NARAYANASWAMI SHRI P. H. CHAUDHARI (<i>Alternate</i>)

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<i>Organization</i>	<i>Representative(s)</i>
Kolkata Municipal Corporation, Kolkata	SHRI D. K. SANYAL SHRI A. K. BISWAS (<i>Alternate</i>)
Mahindra EPC Irrigation Ltd, Nashik	SHRI SANKAR KUMAR MAITI SHRI C. V. JOSHI (<i>Alternate</i>)
Military Engineer Services, Engineer-in-Chief's Branch, Integrated HQ of MoD (Army), New Delhi	SHRI A. K. DUBEY SHRI R. K. CHAUHAN (<i>Alternate</i>)
Ministry of Drinking Water and Sanitation, New Delhi	SHRI DINESH CHAND SHRI SUMIT PRIYADARSHI (<i>Alternate</i>)
NSF Safety and Certification India Pvt Ltd, Gurugram	SHRI B. B. SINGH SHRI NASRIN KASHEFI (<i>Alternate</i>)
Odisha PVC Pipes Manufacturing Association, Bhubaneswar	SHRI NIKUNJ CHHOTRAY
Panchayati Raj and Drinking Water Department, Govt of Odisha, Bhubaneswar	CHIEF ENGINEER
Plastindia Foundation, Mumbai	SHRI RAJIV J. RAVAL DR E. SUNDARESAN (<i>Alternate</i>)
Public Health Engineering Department, Jaipur	SUPERINTENDING ENGINEER EXECUTIVE ENGINEER (<i>Alternate</i>)
Reliance Industries Limited, Mumbai	SHRI S. V. RAJU SHRI SAURABH BAGHAL (<i>Alternate</i>)
RITEs Limited, New Delhi	SHRI PANKAJ AGGARWAL SHRI MUKESH SINHA (<i>Alternate</i>)
Shaktiman Extrusions Pvt Ltd, Perumbavoor	SHRI N. SURESH SHRI T. S. MANOJ (<i>Alternate</i>)
Supreme Industries Limited, Jalgaon	SHRI G. K. SAXENA
Tamil Nadu Water Supply and Drainage Board, Chennai	ENGINEERING DIRECTOR JOINT CHIEF ENGINEER (COM) (<i>Alternate</i>)
Tata Consulting Engineers Ltd, Mumbai	REPRESENTATIVE
In personal capacity (<i>L-202 Metrozone, Anna Nagar West, Chennai 600040</i>)	SHRI G. K. SRINIVASAN
In personal capacity (<i>A-59, Sector 35, Noida 201301</i>)	SHRI KANWAR A. SINGH
BIS Directorate General	SHRI SANJAY PANT, SCIENTIST 'F' AND HEAD (CIVIL ENGINEERING) [REPRESENTING DIRECTOR GENERAL (<i>Ex-officio</i>)]

Member Secretary

SHRIMATI MADHURIMA MADHAV
SCIENTIST 'D' (CIVIL ENGINEERING), BIS

PVC and ABS Piping System Subcommittee, CED 50:2

<i>Organization</i>	<i>Representative(s)</i>
In Personal Capacity (L-202, Metrozone, Anna Nagar West, Chennai 600040)	SHRI G. K. SRINIVASAN (Convener)
Ashirvad Pipes Pvt Ltd, Bengaluru	SHRI MOHAMMAD NOUFAL M. SHRI R. SIVAKUMAR (<i>Alternate</i>)
Astral Poly Technik Ltd, Ahmedabad	SHRI SANDEEP ENGINEER SHRI LALIT TRIVEDI (<i>Alternate</i>)
Baerlocher India Additives Pvt Ltd, Mumbai	DR SHREEKANT DIWAN SHRI SACHIN BIDKAR (<i>Alternate</i>)
Central Ground Water Board, Faridabad	SHRI D. N. ARUN SHRI K. R. BISWAS (<i>Alternate</i>)
Central Institute of Plastic Engineering & Technology, Chennai	SHRI M. NAVANEETHAN
Central Public Works Department, New Delhi	CHIEF ENGINEER (CSQ) EXECUTIVE ENGINEER (S & S) (<i>Alternate</i>)
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Tamil Nadu Water Supply & Drainage Board, Chennai	ENGINEERING DIRECTOR JOINT CHIEF ENGINEER (COM) (<i>Alternate</i>)
In personal capacity (A-59, Sector 35, Noida 201301)	SHRI KANWAR A. SINGH

IS 4985 : 2021

Panel for Revision of IS 4985, CED 50 : P5

<i>Organization</i>	<i>Representative(s)</i>
In personal capacity (<i>L-202, Metrozone Anna Nagar West, Chennai 600040</i>)	SHRI G. K. SRINIVASAN (<i>Convener</i>)
Central Institute of Plastic Engineering and Technology, Chennai	DR S. N. YADAV
Finolex Industries Limited, Pune	SHRI D. J. SALUNKE
Jain Irrigation System Limited, Jalgaon	SHRI M. R. KHARUL
Reliance Industries Ltd, Mumbai	SHRI S. V. RAJU
Shaktiman Extrusions Pvt Ltd, Perumbavoor	SHRI N. SURESH
Supreme Industries Ltd, Jalgaon	SHRI G. K. SAXENA

(Continued from second cover)

ISO/DIS 4422-2-1996	Pipes and fittings made of unplasticized PVC for water supply — Specifications
ISO 9852 : 1995(E)	Unplasticized PVC pipes — Dichloromethane resistance at specified temperature
ISO/DIS 12162	Thermoplastics materials for pipes and fittings for pressure applications — Classification and designation — Overall service (design) coefficient
WHO, Geneva, 1984	Guidelines for drinking water: Recommendations
prEN 1452-1 : 1994	Plastics piping systems for water supply. Unplasticized polyvinyl chloride
prEN 1452-2 : 1994	(PVC-U) — Part 1: General and: Part 2 Pipes
Drafts	

This revision of the standard has been taken up in view of the *Ministry of Environment, Forest and Climate Change Draft Rules*, 2019 dated 12 September 2019 as per which use of lead stabilizers in manufacturing of PVC pipes and fittings has to be phased out. Different timelines for phasing out has been stipulated for pipes for different application. Use of lead as stabilizers in manufacturing of PVC pipes and fittings for water supply will be phased out within one year from the effective date in the notification, whereas that from pipes for agricultural use and sewerage and drainage will be phased out in three years and four years, respectively. The standard covered PVC pipes for both potable water supply and agricultural use which was permitted through a reference in the foreword of the standard. Considering the request received from the industry, the Committee deliberated on the proposed Draft Rules and decided to cover PVC pipes for water supply and agricultural use as Type A and Type B pipes, respectively so that the requirement of lead phase out could be rationally applied.

The following modifications have been incorporated in this revision:

- a) Title of the standard has been modified to ‘Unplasticized PVC pipes for water supplies — Specification’ from ‘Unplasticized PVC pipes for potable water supplies — Specification’.
- b) Use relating to agriculture which was earlier permitted through a reference in the foreword of the standard, has been added in the scope. Further, PVC pipes have been classified based on application, that is, pipes for potable water supply and that for agriculture use.
- c) Marking requirement for pipe for potable water supply has been updated.

The Committee has decided to review from time to time the provisions of this standard with regard to lead phase out to bring the same in accordance with the MoEF & CC notification.

For guidelines on methods of laying and jointing of UPVC pipe work system, including storage, reference may be made to IS 7634 (Part 3) : 1975 ‘Code of practice for plastic pipe work for potable water supplies: Part 3 Laying and jointing of UPVC pipes’.

This standard contains **7.1.4.1** which permits the purchaser to use his option for selection to suit his requirements at the time of placing orders.

The composition of the Committee responsible for the formulation of this standard is given in Annex E.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 1960 ‘Rules for rounding off numerical values (revised)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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